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Effect of sulfur on α-Al_2O_3-supported iron catalyst for Fischer–Tropsch synthesis

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ABSTRACT

The effects of sulfur modification (S/Fe = 0.001 ~ 0.05, molar ratio) on the α-Al_2O_3-supported iron catalysts for the Fischer–Tropsch synthesis (FTS) were investigated. It was found that the presence of sulfur promoted the reduction of FeO to metallic Fe during the H_2 reduction process, but decreased the rates of reduction, carburization and carbon deposition of the catalysts when CO is used as the reducing agent. Both the sulfide (S^{2-}) and sulfate species (SO_4^{2-}) are found in the S_xFe/α-Al_2O_3 catalysts after H_2 reduction and FTS reaction. The sulfide species in the H_2-reduced S_xFe/α-Al_2O_3 catalysts are located on the surface of metallic Fe particles. The intensity of H_2 desorption peak around 500 K is enhanced by a small amount of sulfur, while an excess of sulfur suppresses the H_2 adsorption. The dissociative adsorption of CO on H_2-reduced catalysts was also found to be inhibited by the presence of sulfur. The activity of the Fe/α-Al_2O_3 catalyst can be decreased significantly by the addition of sulfur due to the inhibition of CO dissociation and thus reduce the formation of iron carbide phases under FTS reaction conditions. Resistance to sulfur poisoning of the Fe/α-Al_2O_3 catalyst was found to improve with increasing the reaction temperature. The presence of sulfur also suppressed the formation of C_{5+} hydrocarbons and shifted the products to C_2–C_4 hydrocarbons. At the same time, the olefin to paraffin (O/P) ratio of the C_2–C_4 hydrocarbons decreased with increasing S/Fe molar ratio. These may have resulted from the increasing of the H/C ratio on the surface of sulfur modified catalyst under FTS conditions.

1. Introduction

Fischer–Tropsch synthesis (FTS) is a crucial step in the production of clean fuels and valuable chemicals from syngas (a mixture of H_2 and CO). Syngas can be derived from various non-petroleum carbon resources, such as coal, natural gas, and biomass [1,2]. Compared with other metal catalysts, iron-based catalysts produce less methane and more olefins when the reaction is performed at temperatures higher than 573 K. Additionally, iron-based catalysts have a higher water–gas shift (WGS) activity and a higher resistance to the impurities present in syngas than the other metal catalysts, and are suitable for the CO-rich syngas derived from coal or biomass [2,3].

It is well known that the poisoning of iron-based catalysts by sulfur compounds is a critical problem in FTS to hydrocarbons from coal-derived syngas [4]. Substantial works have been carried out in the past to establish the effects of sulfur on iron-based catalysts [4–22]. However, there are also considerable disagreements and discrepancies among the studies regarding the role of sulfur in the FTS process. In many cases, sulfur will lead to a decline in activity [9,11,12,19] and olefin selectivity of the catalysts [15,20] accompanied by increasing of the selectivity of CH_4 and gaseous C_2–C_4 hydrocarbons [12,15]. On the contrary, some researchers reported that low concentrations of sulfur have positive effects on the activity [13,15,16,18] and olefin selectivity [16] of iron-based FT catalysts. It was also reported that sulfur could improve the heavier hydrocarbon selectivity [21] and reduce CH_4 production in the FTS reaction [16,22]. The effect of sulfur in the FTS process have been reported by Madon et al. [4] and McCue et al. [17] in two excellent reviews. They emphasized that the interactions of sulfur with FTS catalysts appeared to be quite complicated and that both poisoning and promoting effects had been observed. The controversies

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may have resulted from the differences in catalyst composition, the FTS reaction conditions as well as quantity and nature of the sulfur species in the catalysts. In addition to these factors, many studies on the interaction of sulfur with iron catalysts have focused on the exposure of a catalyst to a gaseous sulfur source. This would produce concentration gradients of sulfur along the length of the catalyst bed and pose severe problems in the interpretation of the effect of sulfur. At the same time, interpretation of the effect of sulfur in a catalyst can be clouded by other modifiers in the catalysts such as alkali metals (Na [13,16,23], K [5,7,15,19]). In spite of the recognized importance of the sulfur, the roles of sulfur in FTS reaction is still not well understood and remains to be explored. For this reason, a detailed study of the effect of sulfur on the iron catalysts without any other modifiers was carried out. Particular attention has been paid to surface compositions and nature of the sulfur species in the freshly prepared, reduced and used catalysts as well as the effect of sulfur on the textural, reduction and carburization properties of the catalysts. Sulfur-modified iron catalysts were prepared via impregnation of the support with a solution of iron nitrate and ammonium sulfate. Such a method allows us to get a better control over the quantity of sulfur introduced as a well as a more uniform distribution of sulfur species in the catalyst bed. α-Al_2O_3 was chosen because it is an excellent support for iron catalyst in the selective production of lower olefins from syngas and allows a better interaction between the sulfur and the iron phase [16,24,25]. Catalyst with S/Fe molar ratios varying from 0.001 to 0.05 were prepared and were characterized by transmission electron microscopy (TEM), (*in situ*) X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and temperature-programmed reduction/desorption (TPR/TPD), whereas the FTS reaction is conducted in a fixed-bed reactor at industrially relevant operating conditions. It is expected that the relevant studies can provide a better understanding of the nature of the interactions between sulfur and iron, and will be helpful for predicting lifetime as well as product distribution of the catalyst.

2. Experimental

2.1. Preparation of sulfur-modified Fe/α-Al_2O_3 catalysts

The sulfur-modified Fe/α-Al_2O_3 catalysts were prepared via the incipient wetness impregnation method. 5.0 g of α-Al_2O_3 (surface area: 8 m^2/g, pore volume: 0.24 mL/g, Alfa Aesar) was impregnated with an aqueous solution (1.5 mL) of $Fe(NO_3)_3\cdot 9H_2O$ (3.5 g, Sinopharm Chemical Reagent, ≥98.5 %) and $(NH_4)_2SO_4$ (0.0016 ~ 0.0817 g, Sinopharm Chemical Reagent, ≥99.0 %). The α-Al_2O_3 support was treated in vacuum at 383 K for 12 h before use. The samples were then dried at 383 K for 12 h. The solid materials were finally heated under an air flow of 30 mL/min to 673 K (heating rate = 10 K/min) and calcined at 673 K for 2 h to obtain the oxidized catalyst precursors denoted as S_xFe/α-Al_2O_3, in which x represents S/Fe molar ratio. The Fe loading in all catalysts was 10 wt%. The α-Al_2O_3 support and Fe/α-Al_2O_3 catalyst were tested by a Bruker Tiger S8 X-ray fluorescence (XRF) spectrometer. The results showed that content of sodium and sulfur in the samples was below the instrument detection limit.

2.2. Catalyst characterizations

The microstructure of Fe/α-Al_2O_3 catalysts was examined *ex-situ* by a FEI Tecnai F30 transmission electron microscopy (TEM) operated at an accelerating voltage of 300 kV. The sample for TEM measurement was prepared by depositing a drop of Fe/α-Al_2O_3/ethanol suspension on a carbon-film-coated copper grid.

X-ray powder diffraction (XRD) analysis was carried out by a Panalytical X'pert PRO diffractometer scanning 2θ from 20 to 80°. Cu-Kα radiation (λ = 1.5406 Å) obtained at 40 kV and 30 mA was used as the X-ray source. The *in situ* XRD experiment was performed on the same instrument with an Anton Paar XRK-900 reaction chamber. In the *in situ* XRD experiment, diffraction pattern of the catalyst was collected with 2θ ranging from 20 to 80°. The scanning rate was 3 °/min. In each *in situ* XRD experiment, the sample was heated in flowing H_2 (99.999%, 120 mL/min) up to 873 K at a rate of 10 K/min and the XRD patterns were recorded at the prespecified temperatures.

H_2 temperature-programmed reduction (H_2-TPR) experiment was performed on a GC-TPR apparatus. Prior to the TPR test, the fresh catalyst (0.10 g) was pretreated with a flow of O_2/Ar = 5/95 mixture (v/v, 30 mL/min) at 573 K for 30 min. After cooled down to 323 K, the sample was purged with a flow of H_2/Ar = 5/95 mixture (v/v, 50 mL/min) until the baseline was flat. The TPR profile of the catalyst was obtained by heating the sample from 323 to 1173 K at a rate of 10 K/min. The hydrogen consumption was monitored by a TCD.

CO temperature-programmed reduction (CO-TPR) experiment was performed on a TPR apparatus equipped with a Hiden Qic-20 mass spectrometer (MS). Typically, the fresh catalyst (0.10 g) was pretreated with a flow of He (99.999 %, 30 mL/min) at 673 K for 15 min and cooled down to 373 K. The sample was then purged with a CO/He = 5/95 mixture (v/v, 50 mL/min) at 373 K until the baseline was flat. CO-TPR was performed in the CO/He flow (50 mL/min) by raising the temperature to 1073 K at a rate of 10 K/min. Ions with *m/z* values at 28 (CO) and 44 (CO_2) were detected during on-line measurements.

H_2 temperature-programmed desorption (H_2-TPD) experiment was performed on a TPD apparatus equipped with a Hiden Qic-20 mass spectrometer. The catalyst (0.10 g) was first reduced with a flow of pure H_2 (30 mL/min) at 673 K for 120 min. After cooled to 303 K under H_2, the sample was purged with Ar (99.999%, 50 mL/min) at 303 K for about 30 min until the baseline leveled off. The H_2-TPD profile of the catalyst was obtained by heating the sample from 323 to 1073 K at a rate of 10 K/min in a flow of Ar (50 mL/min). Ions with *m/z* values at 2 (H_2) and 18 (H_2O) were detected during the on-line measurements.

The CO-TPD experiment was performed on the same apparatus as used in the H_2-TPD experiment. The fresh catalyst (0.10 g) was first reduced with pure H_2 (99.999%, 30 mL/min) at 673 K for 2 h. The sample was then flushed with He flow (99.999%, 50 mL/min) at 673 K for 10 min and cooled down under He to 303 K. CO adsorption was performed by switching the He flow to a CO flow (99.99 %, 30 mL/min) at 303 K and treating for 10 min. The sample was then purged with He flow for about 30 min until the baseline was flat. The CO-TPD profile of the catalyst was obtained by heating the sample from 303 to 1073 K at a rate of 10 K/min in a flow of He (50 mL/min). Ions with *m/z* values at 28 (CO) and 44 (CO_2) were detected during the on-line measurement.

X-ray photoelectron spectroscopy (XPS) analysis was performed on an Omicron Sphera II hemispherical electron energy analyzer with an *in situ* reaction cell attached to the instrument. Monochromatic Al Kα X-ray source (1486.6 eV, anode operating at 15 kV and 300 W) was used as incident radiation. The binding energy (BE) of the element was reference to the Al 2p1/2 photoelectron peak at 74.3 eV. This reference gave the BE values with an accuracy of ±0.1 eV. The surface atomic ratio was estimated from the peak areas of the elements corrected by the relative sensitivity factors of the corresponding elements [26]. In each experiment, the catalyst sample in the *in situ* reaction cell was first heated under a flow of O_2/N_2 = 21/79 mixture (v/v, 50 mL/min) from room temperature to 673 K at a heating rate of 10 K/min and treated at 673 K for 20 min. The treated sample was then transferred under vacuum to the analytical chamber to record the XPS spectra of the catalyst in the oxidation state. After that, the catalyst was transferred again to the reaction cell, and heated at a rate of 10 K/min from room temperature to 703 K under a flow of H_2 (50 mL/min) and reduced at 703 K for 120 min. After cooling to 423 K under H_2 flow, the reduced sample was evacuated and transferred under vacuum to the analytical chamber to record the XPS spectra of the catalyst in the reduction state. Finally, the reduced catalyst was transferred to the reaction cell and switched to a flow of H_2/CO = 1 mixture (v/v, 10 mL/min) at 573 K to conduct the CO hydrogenation reaction. After treating with H_2/CO mixture at 573 K for 60 min, the catalyst sample was evacuated and transferred under vacuum to the analytical chamber to record the XPS spectra of the reacted catalyst.

2.3. Catalytic reaction

The Fischer–Tropsch synthesis (FTS) reaction was carried out at 2.0 MPa, 573 K and a H_2/CO ratio of 1 (v/v) in a stainless steel high-pressure fix-bed flow reactor (7.0 mm inner diameter). H_2/CO ratio of 1 is close to that for syngas derived from coal or biomass. Prior to the reaction, the catalyst was reduced by H_2 (0.10 MPa) at 673 K for 2 h (heating rate = 10 K/min). After the reduced catalyst was cooled to 473 K, a syngas flow of H_2/CO/Ar = 48/48/4 (v/v/v) was introduced into the reactor. The pressure in the reactor was then increased gradually from 0.10 to 2.0 MPa and the temperature was raised to the desired temperature. The products were analyzed by two on-line gas chromatographs. Ar, CO, CO_2, CH_4, C_2H_4, C_2H_6, C_3H_6 and C_3H_8 were analyzed by a SICT (Shanghai Institute of Computing Technology) GC-2000 III gas chromatograph equipped with a TCD, and a TDX-02 and a Porapak Q columns using H_2 as a carrier gas. The hydrocarbons were analyzed by a SICT GC-2000 III equipped with a FID and a PONA capillary column using He as a carrier gas. 4% Ar was used as an internal standard for calculation of CO conversion. The product selectivity was determined by a methane concentration correlation method and was calculated on a carbon basis.

3. Results and discussion

The XRD patterns of the fresh S_xFe/α-Al_2O_3 catalysts with S/Fe atomic ratios between 0 and 0.05 are shown in Fig. 1. Besides diffraction peaks of α-Al_2O_3, the diffraction peaks originating from iron oxide with 2θ = 24.2, 33.2, 35.6, 40.9, 49.5, 54.1, 62.5, 64.0° are also detected. The latter are similar to that of the hematite (α-Fe_2O_3) species (JCPDS No. 01-084-0306). However, no diffraction peak belonging to a sulfur compound is seen. This may have

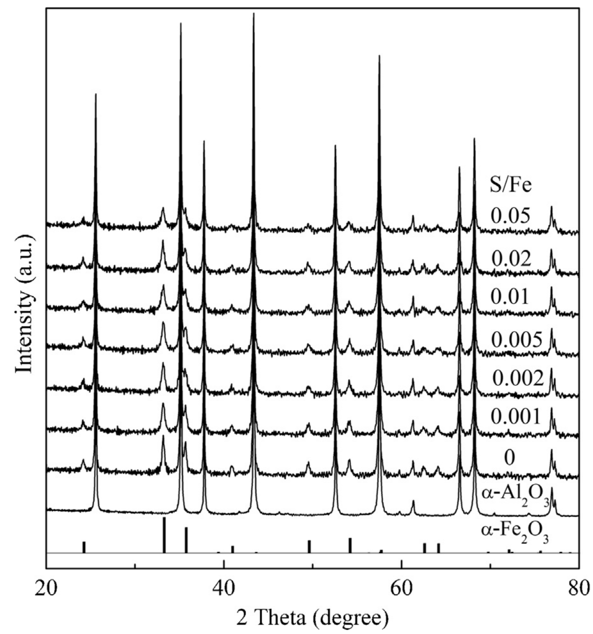

Fig. 1. XRD patterns of the as prepared S_xFe/α-Al_2O_3 catalysts with different S/Fe molar ratios.

resulted from that the sulfur content in the catalysts is below the detection limit of XRD. With the increasing of sulfur content in the catalysts, the intensity of the characteristic peaks of α-Fe_2O_3 in the catalysts has been gradually weakened. This effect may be attributed to less of crystallinity as a result of the interaction between Fe_2O_3 and sulfur.

The morphology and size of the iron species in the Fe/α-Al_2O_3 and $S_{0.01}Fe/\alpha$-Al_2O_3 catalysts after H_2 reduction at 673 K was studied by TEM. As shown in Fig. 2, the iron-containing nanoparticles located at the exterior surface of α-Al_2O_3 can be easily distinguished from the support material. Average iron (oxide) particle sizes of the Fe/α-Al_2O_3 and $S_{0.01}Fe/\alpha$-Al_2O_3 catalysts are 15 ± 5 nm and 16 ± 6 nm, respectively. Both catalysts showed similar morphology.

H_2-TPR and *in situ* XRD experiments were carried out to study the effect of sulfur on the reducibility of the S_xFe/α-Al_2O_3 catalysts. As shown in Fig. 3, the H_2-TPR profile of the sulfur-free Fe/α-Al_2O_3 catalyst showed three distinct reduction stages in the temperature range of 500–1100 K with temperature at peak maximum (T_m) at

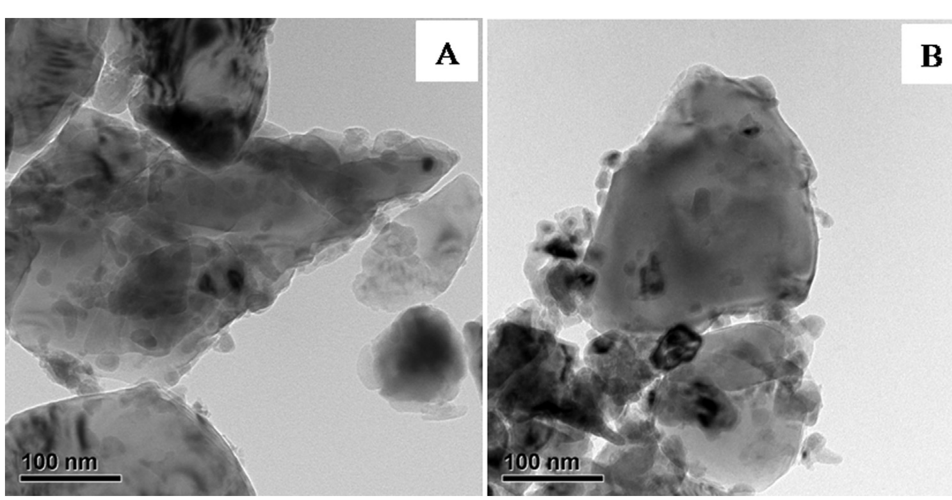

Fig. 2. TEM images of the (A) Fe/α-Al_2O_3 and (B) $S_{0.01}Fe/\alpha$-Al_2O_3 catalysts after H_2 reduction at 673 K for 2 h.

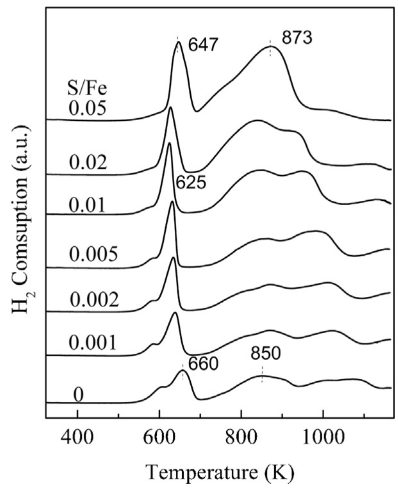

Fig. 3. H_2-TPR profiles of the S_xFe/α-Al_2O_3 catalysts with different S/Fe molar ratios.

660 K (peak 1), ~850 K (peak 2) and above 940 K (peak 3). From the deconvolution results of the TPR profile of the Fe/α-Al_2O_3 catalyst (Table S1, Supporting information), it can be seen that the ratio of the peak areas for peak 1 and peak 2 is close to the theoretical ratio of the H_2 consumption corresponding to the reduction of Fe_2O_3 to Fe_3O_4 and Fe_3O_4 to FeO, respectively. The peak with T_m above 940 K (peak 3) can be assigned to the reduction of FeO to metallic Fe. Compared with the areas of peak 1 and peak 2, the area of peak 3 (at temperatures below 1073 K) is lower than that expected for the complete reduction of FeO to metallic Fe. This may have resulted from the formation of unreducible iron aluminate as a result of strong interaction between FeO and the alumina support. When sulfur was added, the T_m for the reduction of Fe_2O_3 to Fe_3O_4 decreased from 660 to 640 K, and a further shift of T_m to lower temperature was observed as S/Fe molar ratio increased from 0.001 to 0.01. Similar phenomenon was observed in the H_2-TPR profiles of the unsulfided and sulfide iron catalysts [13]. As shown in Fig. 3, with the increasing of S/Fe molar ratio in the catalysts from 0 to 0.05, the T_m of peak 3 also shifted towards lower temperature. This result suggests that the presence of sulfur also promoted the reduction of FeO to metallic Fe in H_2. A similar result was reported when sulfur modified Fe-based catalysts were reduced in H_2 [13,15]. According to the study of Li et al. [15], the higher reducibility of sulfur-modified iron-based catalysts was attributed to the decreasing of catalyst surface basicity in the presence of sulfur. Sulfur was also found to assist with oxygen removal from the iron-oxide lattice [13], and may increase the catalyst reducibility.

The XRD patterns of Fe/α-Al_2O_3 catalyst recorded in a flow of H_2 (120 mL/min) during stepwise heating of the sample from 298 to 873 K are shown in Fig. 4A. When the temperature of the sample was raised to 593 K, diffraction peaks of Fe_2O_3 (2θ = 24.2, 33.2, 35.6, 40.9, 49.5, 54.1, 62.5, 64.0°) decreased significantly in intensity, and the characteristic peaks of Fe_3O_4 phase (2θ = 30.1, 35.4, 43.1, 56.9, 62.5°) were observed, indicating the reduction of Fe_2O_3 to Fe_3O_4. The XRD peaks of Fe_2O_3 were found to vanish when temperature was raised to 613 K. The diffraction peaks attributed to FeO (2θ = 36.1, 41.9°) and metallic Fe (2θ = 44.7, 65.0°) started to emerge at 653 K, revealing the transformation of Fe_3O_4 to FeO and metallic Fe. When temperature was increased to 713 K, the diffraction peaks of Fe_3O_4 disappeared and the intensity of the FeO peak went to the maximum. With a further rise in the temperature, the peak intensity of metallic Fe increased gradually. The reduction behavior of iron-based catalysts has also been investigated extensively by the Mössbauer spectroscopy [27–31]. Based on the results of *in situ* Mössbauer spectroscopy characterization on the phase transformation of a precipitated iron-based catalyst during isothermal reduction, Wang et al. found that the reduction of paramagnetic (PM) α-Fe_2O_3 (70%) and superparamagnetic (spm) Fe^{3+} (30%) in the fresh catalyst proceed via different steps. PM α-Fe_2O_3 is firstly reduced to magnetite and then to metallic iron, while the reduction of spm Fe^{3+} proceeds in three consecutive steps: it is first reduced to magnetite with a significantly rapid rate, then to nonstoichiometric wüstite, and finally to metallic iron [31]. In general, these results is consistent with the results of *in situ* XRD characterization on the Fe/α-Al_2O_3 in the present study which indicate that reduction of the Fe_2O_3 species on α-Al_2O_3 in H_2 to metallic Fe is a three-step process (*i.e.* $Fe_2O_3 \rightarrow Fe_3O_4 \rightarrow$ FeO → metallic Fe).

Similar XRD patterns were observed when a $S_{0.02}Fe/\alpha$-Al_2O_3 catalyst was heated in a H_2 flow from 473 to 873 K (Fig. 4B). However, compared to the reduction of Fe/α-Al_2O_3 catalyst, the

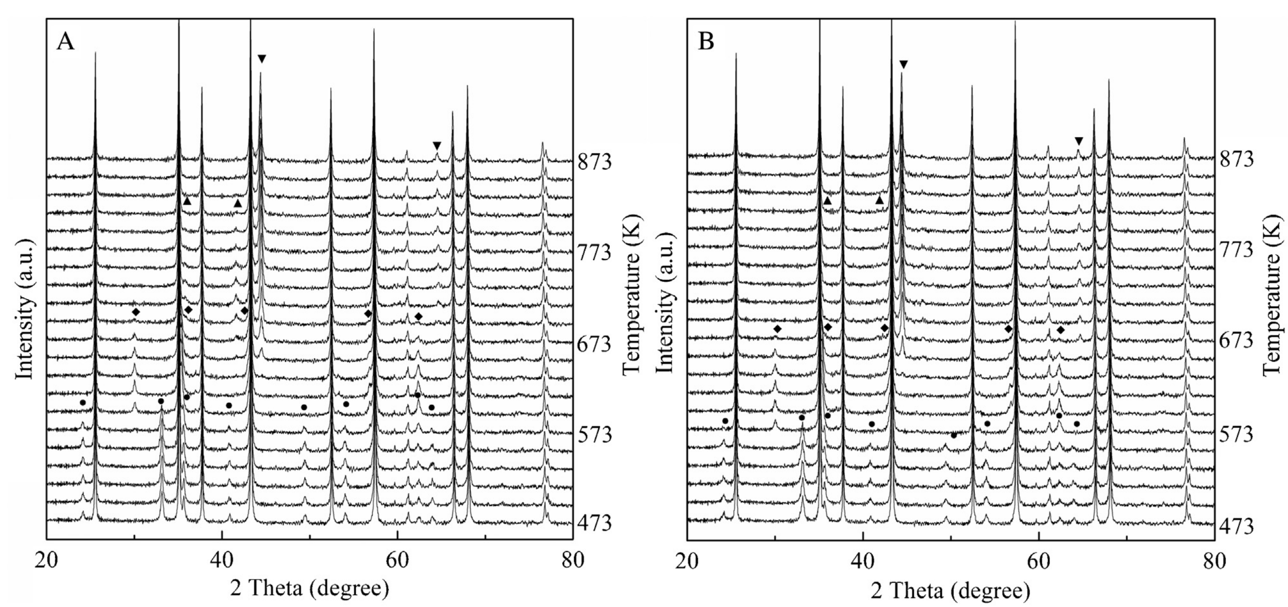

Fig. 4. *In situ* XRD patterns of (A) Fe/α-Al_2O_3 and (B) $S_{0.02}Fe/\alpha$-Al_2O_3 catalyst recorded at the indicated temperature during H_2 reduction process. (● α-Fe_2O_3 ♦ Fe_3O_4 ▲ FeO ▼ Fe).

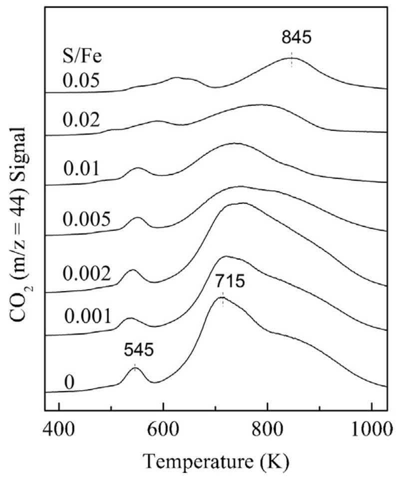

Fig. 5. CO-TPR profiles for the $S_xFe/\alpha\text{-}Al_2O_3$ catalysts with different S/Fe molar ratios.

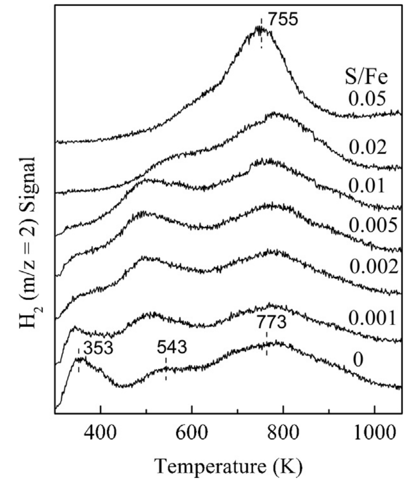

Fig. 6. H_2-TPD profiles of the H_2-reduced $S_xFe/\alpha\text{-}Al_2O_3$ catalysts with different S/Fe molar ratios.

reduction of Fe_2O_3 to Fe_3O_4 in $S_{0.02}Fe/\alpha\text{-}Al_2O_3$ occurred at a lower temperature (573 K), and more metallic Fe species were formed in the $S_{0.02}Fe/\alpha\text{-}Al_2O_3$ catalyst when reduction temperature was raised to 653 K. Another noticeable difference between the XRD pattern of $Fe/\alpha\text{-}Al_2O_3$ and $S_{0.02}Fe/\alpha\text{-}Al_2O_3$ catalysts is that no diffraction peak corresponding to FeO was detected in the latter during the reduction process. These results are in agreement with the results of H_2-TPR characterizations shown in Fig. 3, and indicate that the presence of sulfur promotes the reduction of Fe_2O_3 to Fe_3O_4 as well as the reduction of FeO to metallic Fe.

CO-TPR experiments were carried out to investigate the effect of sulfur on the reduction and carburization of the Fe species in the $S_xFe/\alpha\text{-}Al_2O_3$ catalysts under CO atmosphere and the results are shown in Fig. 5. Two groups of CO_2 peaks are seen in the CO-TPR profile of sulfur-free $Fe/\alpha\text{-}Al_2O_3$ catalyst. The first CO_2 formation peak observed in the temperature range of 450–570 K (peak 1) can be assigned to the reduction of Fe_2O_3 to Fe_3O_4. Fe_3O_4 is the only Fe containing species detected by XRD in the catalyst sample after CO reduction at 573 K (Fig. S1). The CO_2 formation peak above 570 K (peak 2) can be related to the conversion of CO via the Boudouard reaction ($2CO \rightarrow C + CO_2$) as well as the reduction of Fe_3O_4 to metallic Fe and carburization of Fe_3O_4 to the mixtures of iron carbides [32]. The carburization of Fe_3O_4 to iron carbides is evidenced by the observation of a mixture of iron carbides in the XRD pattern of the sulfur-free $Fe/\alpha\text{-}Al_2O_3$ catalyst after the CO reduction at 1073 K (Fig. S1). The assignment of peak 2 is also supported by the peak deconvolution results of the CO-TPR signals of the sulfur-free $Fe/\alpha\text{-}Al_2O_3$ catalyst (Table S2). It can be seen that the intensity ratio (1/12) of peak 1 and peak 2 in Fig. 5 is smaller than the stoichiometry (1/8) of CO_2 production for the reduction of Fe_2O_3 to Fe_3O_4 and Fe_3O_4 to Fe by CO, suggesting that other reactions such as carburization of the iron species (Fe_3O_4 or Fe) to iron carbides and the Boudouard reaction also took place during the CO-TPR process. According to the results of Mössbauer analysis [33], a Mo-promoted Fe catalysts (100Fe5Mo) pretreated with pure CO at 280 °C were reduced to Fe_3O_4, $\chi\text{-}Fe_5C_2$ and superparamagnetic species (Fe^{3+} and/or Fe^{2+}). The results of *in situ* TPR-EXAFS/XANES study also confirmed that the Fe/Si catalysts (100Fe:4.6Si) promoted with group I alkali metals were firstly reduced to Fe_3O_4 and then carburized to iron carbides under a CO/He flow [34]. In summary, The Fe_2O_3 species in the $Fe/\alpha\text{-}Al_2O_3$ catalyst undergoes a consecutive stepwise reduction to Fe_3O_4 and iron carbides under CO atmosphere.

With the increasing of S/Fe molar ratio in the $S_xFe/\alpha\text{-}Al_2O_3$ catalysts from 0.002 to 0.05, the intensity of the second CO_2 peak in Fig. 5 decreases gradually. It was also found that the temperature of the two CO_2 formation peaks increases significantly with the increasing of S/Fe molar ratio from 0.01 to 0.05. Moreover, the result of XRD measurements (Fig. S1) indicated that only a metallic Fe phase was detected in the $S_{0.05}Fe/\alpha\text{-}Al_2O_3$ sample reduced with CO at 1073 K. These results suggest that the addition of sulfur decreases the rates of reduction and carburization of the iron species in $S_xFe/\alpha\text{-}Al_2O_3$ catalysts by CO. Similar results were reported by Li et al. [15] in the study of sulfate promoted iron-manganese catalysts. Obviously, the reduction behavior of the $S_xFe/\alpha\text{-}Al_2O_3$ catalysts by CO (Fig. 5) is different from that by H_2 (Figs. 3 and 4). This may have resulted from that CO dissociation on the metallic iron surface is retarded by the presence of sulfur (as observed in CO-TPD experiment shown below) [35–37]. It was reported that the presence of sulfur on the iron surface results in the depletion of electrons from the iron [35] and/or blocking of the available CO dissociation sites and decreasing the CO dissociative adsorption [36,37]. As a result of these, the reduction, carburization and carbon deposition of the iron species in the catalysts under CO atmosphere are inhibited.

The H_2-TPD was used to investigate the effect of sulfur on the H_2 adsorption behavior of the catalysts. As can be seen from Fig. 6, three peaks with temperature at peak maxima (T_m) at 353, 543 and 773 K were seen in the H_2-TPD profile of $Fe/\alpha\text{-}Al_2O_3$ catalyst. As reported in literature, hydrogen desorption usually occurs below 600 K on single-crystal iron planes [38,39] or polycrystalline iron surfaces [40–43], which is similar to the low temperature desorption peaks observed on the $Fe/\alpha\text{-}Al_2O_3$ catalyst in the present study. The H_2 desorption peak at temperatures above 600 K can be ascribed to a strong chemisorption state of hydrogen on the catalyst [44]. It was also reported that water held by the Al_2O_3 support could be driven off at temperatures above 713 K and reacted with the metallic Fe to form a H_2 peak at above 700 K [40]. Nevertheless, it was found that the hydrogen species in a strong chemisorbed state could not react with N_2 to form ammonia at temperature below 773 K [44]. Considering the CO hydrogenation temperature in this study is below 623 K, it is rational to suggest that the hydrogen species desorbed from the $Fe/\alpha\text{-}Al_2O_3$ catalyst at temperature below 623 K (Fig. 6) should be responsible for the CO hydrogenation reaction over the $Fe/\alpha\text{-}Al_2O_3$ catalyst.

As illustrated in Fig. 6, the H_2 desorption peak at 353 K decreased gradually in intensity with the increasing of S/Fe molar ratio in the catalysts from 0 to 0.01, and eventually disappeared at S/Fe = 0.02. Whereas the intensity of H_2 desorption peaks at 543 K increased with the increasing of S/Fe molar ratio from 0.001 to 0.002 and remained unchanged with a further increase in the S/Fe molar ratio

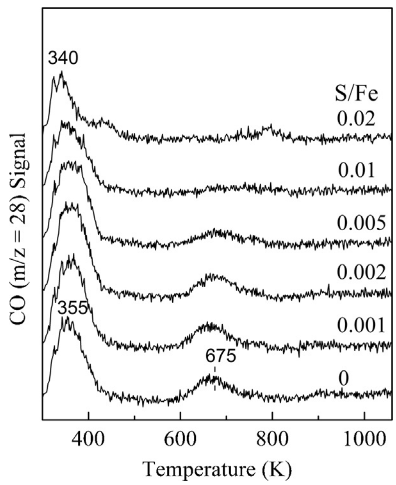

Fig. 7. CO-TPD profiles of the S_xFe/α-Al_2O_3 catalysts with different S/Fe molar ratios.

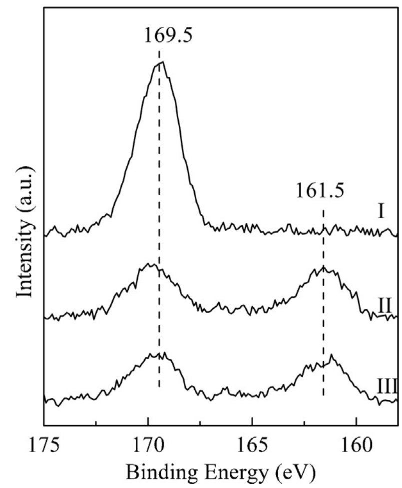

Fig. 8. S 2p XPS spectra of the $S_{0.05}Fe/\alpha$-Al_2O_3 catalyst after (I) treating under a flow of O_2/N_2 = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) reducing under a flow of H_2 (50 mL/min) at 703 K for 120 min, (III) exposure to a flow of H_2/CO = 1 mixture (v/v, 10 mL/min) at 573 K for 60 min.

from 0.005 to 0.01. Similar phenomena were observed on the industrial iron catalysts poisoned with different amounts of sulfur [45]. These results suggest that a small amount of sulfur can enhance the amount of H_2 desorption from iron catalyst surface at about 500 K. The underlying reason for the observation may be that sulfur is conducive to the reduction of Fe_2O_3 to Fe^0 in H_2, and formation of metallic Fe species is in favor of the activation and adsorption of hydrogen on the catalyst. On the $S_{0.05}Fe/\alpha$-Al_2O_3 catalyst, the H_2 desorption peaks at temperature below 600 K almost completely disappeared. This phenomenon can be understood in terms of blockage of the site for H_2 adsorption on the catalyst surface by sulfur at relatively high sulfur coverage. It was found that sulfur adsorption into $c(2 \times 2)$ structures on Fe(100) surface completely blocked the site for H_2 adsorption at 200 K [35].

Several studies have been carried out to study the effect of sulfur on the adsorption behavior of CO over low index planes of iron metal surface [35,36,46–48], whereas the effect of sulfur on CO adsorption over supported iron catalysts is rarely studied. To shed more light on the effect of sulfur on CO adsorption and dissociation over α-Al_2O_3 supported iron catalysts, a CO-TPD study of the H_2-reduced S_xFe/α-Al_2O_3 catalysts was conducted. The results are shown in Fig. 7. The adsorption of CO on H_2-reduced Fe/α-Al_2O_3 catalyst resulted in two major CO desorption peaks in the temperature range of 300 and 1050 K. The peak at lower temperature (355 K) corresponds to the desorption of molecularly adsorbed CO, while the one at higher temperature (675 K) is ascribed to recombination of C and O of the dissociated CO [49,50]. These results are in agreement with the CO desorption profiles on Fe/SiO_2 [49] and Fe/HOPG [50] catalysts. As reported in the literature [35,51], the dissociated carbon and oxygen may have resulted from further dissociation of molecularly absorbed CO at 303 K and higher temperature. As the S/Fe molar ratio increased (especially on the sample with S/Fe molar ratio higher than 0.005), the desorption peak of molecularly adsorbed CO shifted to lower temperature, and the desorption peak of dissociated CO decreased in intensity. On the sample with S/Fe molar ratio higher than 0.01, the desorption peak of dissociated CO disappeared completely. These results clearly showed that the addition of sulfur inhibited the dissociative adsorption of CO on H_2-reduced Fe/α-Al_2O_3 catalysts, which is consistent with the trend observed on the low index planes of iron metal [35,36]. These results can be understood in terms of the bonding interaction between the Fe surface and sulfur atoms which deplete high-lying occupied Fe D-states near E_f and reduce the metal's ability to back-donate electrons into the 2π acceptor orbital of CO [35,52]. In addition, sulfur can also reduce the CO dissociation by blocking the available sites for dissociated carbon and oxygen [36]. It has been found that the dissociated carbon and oxygen atoms occupy randomly the four-fold hollow sites on Fe(100) [53] while sulfur atoms occupy the same sites on Fe(100) [54].

To gain further insight into the nature of the sulfur species in the S_xFe/α-Al_2O_3 catalysts before and after the FTS reaction, a detail XPS analysis on the catalyst was performed. Figs. 8 and 9 show the S 2p and Fe 2p XPS spectra of a freshly prepared $S_{0.05}Fe/\alpha$-Al_2O_3 catalyst and the samples after H_2 reduction at 703 K as well as after the FTS reaction at 573 K. The corresponding surface atomic ratios of the elements over the samples calculated based on the XPS results are given in Table 1. The single peak with B.E. at 169.5 eV observed in S 2p spectrum I indicates that the sulfur is in the form of sulfate (SO_4^{2-}) on the surface of a freshly prepared $S_{0.05}Fe/\alpha$-Al_2O_3 catalyst [23]. It is also found that the S_{total}/Fe ratio on the fresh $S_{0.05}Fe/\alpha$-Al_2O_3 catalyst is much higher than the stoichiometry of S and Fe elements in the catalyst, indicating that the sulfate is mainly located on the catalyst surface. After reducing with H_2 at 703 K for 2 h, XPS peak intensity of sulfate decreased significantly, and a new peak with B.E. at 161.5 eV corresponding to sulfide species (S^{2-}) was observed [23,55], indicating the partial transformation of sulfate species into sulfides after H_2 reduction at 703 K. The calculated surface S^{2-}/S_{total} atomic ratio reveals 45% of S^{2-} in total sulfur, indicating that the sulfide (S^{2-}) and sulfate species (SO_4^{2-}) are the main sulfur components in the H_2-reduced S_xFe/α-Al_2O_3 catalysts. Bromfield et al. studied the effect of reduction temperature on surface sulfur species over the sulfided precipitated-iron catalysts using XPS [23]. It was found that besides assisting in the conversion of sulfate to sulfide, H_2 reduction at high temperature (673 K) also caused segregation of sulfur species on the iron surface, and the diffusion of sulfur species to the surface is favored by higher reduction temperatures. From the results shown in Table 1, how-

Table 1
Surface composition of the $S_{0.05}Fe/\alpha$-Al_2O_3 catalyst after the O_2/N_2 calcination at 673 K, H_2 reduction at 703 K and FTS reaction at 573 K.

Pretreatments	Atomic ratios calculated from XPS				
	Fe/Al	S_{total}/Al[a]	S_{total}/Fe	S^{2-}/Fe[b]	S^{2-}/S_{total}
O_2/N_2, 673 K	0.26	0.052	0.20	0	0
H_2, 703 K	0.19	0.035	0.18	0.083	0.45
H_2/CO, 573 K	0.18	0.034	0.18	0.086	0.47

[a] S_{total}: total amount of sulfur species.
[b] S^{2-}: the amount of sulfide species.

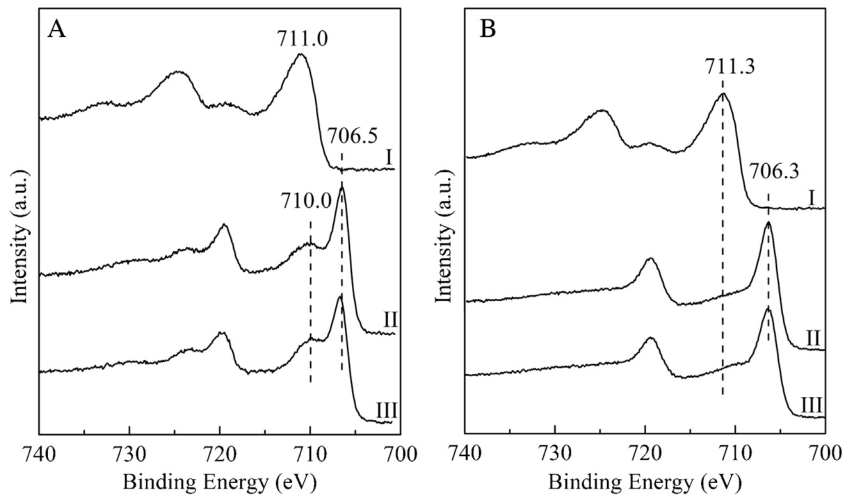

Fig. 9. Fe 2p XPS spectra of the (A) Fe/α-Al_2O_3 and (B) $S_{0.05}$Fe/α-Al_2O_3 catalysts after (I) treating under a flow of O_2/N_2 = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) reducing under a flow of H_2 (50 mL/min) at 703 K for 120 min, (III) exposure to a flow of H_2/CO = 1 mixture (v/v, 10 mL/min) at 573 K for 60 min.

ever, the atomic ratio of S_{total}/Al decreased after the $S_{0.05}$Fe/α-Al_2O_3 catalyst was reduced with H_2 at 703 K. This phenomenon could have resulted from the aggregation of iron and sulfur species during the H_2 reduction process as the atomic ratio of Fe/Al also decreased after H_2 reduction. Another possible reason for the decrease of surface S_{total}/Al ratio in the reduced catalyst is the release of sulfur species as gaseous SO_2 or H_2S during the reduction process. In consideration of the facts that no sulfide species was detected in the S 2p XPS peaks of a α-Al_2O_3 supported sulfate sample ((SO_4^{2-})/α-Al_2O_3) reduced by H_2 at 703 K (Fig. S2), and that only the sulfide species was observed on a precipitated-iron catalyst after exposure to hydrogen at 673 K [23]. It is rational to infer that the sulfide species in the H_2-reduced S_xFe/α-Al_2O_3 catalysts are located on the surface of metallic Fe particles. As can be seen from Fig. 9A, XPS spectrum of a freshly prepared Fe/α-Al_2O_3 shows binding energy of 711.0 eV for the photoelectron peak of Fe $2p_{3/2}$, which is typical for the Fe^{3+} species in the Fe_2O_3 [56,57]. After adding sulfur, the binding energy of Fe $2p_{3/2}$ increases slightly (711.3 eV). This may have resulted from the formation of $Fe_2(SO_4)_3$ as the binding energy of the Fe $2p_{3/2}$ XPS peak for the Fe^{3+} in $Fe_2(SO_4)_3$ (711.3 eV) is higher than that in Fe_2O_3 [58]. After reducing with H_2 at 703 K for 120 min, the XPS peak of Fe^{3+} (711.0 eV) decreased significantly in intensity. A broad peak with binding energy at 710.0 eV and a sharp peak with binding energy at 706.5 eV, assignable to Fe^{2+} [59] and metallic Fe species [57,59,60], respectively, were detected, indicating the reduction of Fe_2O_3 to FeO and Fe^0. Comparing with Fe/α-Al_2O_3, the Fe $2p_{3/2}$ XPS peak for Fe^{3+} in the H_2-reduced $S_{0.05}$Fe/α-Al_2O_3 almost vanished, indicating that most of the Fe^{3+} species on the surface of sulfur modified catalyst was reduced to metallic Fe. This result is also in agreement with the results of H_2-TPR and *in situ* XRD characterizations shown above (Figs. 3 and 4), and provide further evidence to show that sulfur promotes the reduction of Fe_2O_3 to metallic iron.

When H_2-reduced Fe/α-Al_2O_3 sample was exposed to a mixture of CO/H_2 = 1 (v/v) at 573 K for 60 min, the XPS peak of Fe $2p_{3/2}$ at 706.5 eV shifted to 706.7 eV. The latter is in agreement with the Fe $2p_{3/2}$ XPS peak of iron carbide species [57]. This result indicates that metallic iron transforms to iron carbide under the CO/H_2 atmosphere. The formation of iron carbide was further confirmed by the observation of a C 1s peak at 283.0 eV assignable to carbide species [57,59–61] on the CO/H_2 treated Fe/α-Al_2O_3 sample (Fig. S3). Comparing with the CO/H_2 pretreated sulfur free Fe/α-Al_2O_3 catalyst, no significant shift in the Fe $2p_{3/2}$ XPS peak was observed on the CO/H_2 pretreated $S_{0.05}$Fe/α-Al_2O_3 sample, indicating that most of the Fe species remained in metallic state under FTS reaction conditions. This is also confirmed by the C 1s XPS spectra of the sample (Fig. S3) in which no obvious XPS peak corresponding to the carbide species was observed on the CO/H_2 pretreated $S_{0.05}$Fe/α-Al_2O_3. These results indicate that the addition of sulfur suppresses carburization of metallic Fe species.

Comparing with the H_2-reduced $S_{0.05}$Fe/α-Al_2O_3 sample, no significant change in surface atomic ratios and binding energy of the elements was seen on the $S_{0.05}$Fe/α-Al_2O_3 catalyst after the FTS reaction at 573 K for 1 h, indicating that the chemical state and surface concentration of the sulfur species were not significant affected by syngas.

To investigate the effect of sulfur on the activity and selectivity of the α-Al_2O_3 supported iron catalyst for the FTS reaction, CO hydrogenation performance of the S_xFe/α-Al_2O_3 catalysts with different S/Fe molar ratios was tested at 573 K and 2.0 MPa using H_2/CO = 1 as reaction feed, and the results are summarized in Table 2. It can be seen that sulfur modification in a range of S/Fe molar ratio between

Table 2
Catalytic performance of S_xFe/α-Al_2O_3 catalysts for CO hydrogenation.

S/Fe (mol/mol)	Xco (%)	FTY (10^{-4} $mol_{CO}/(s\ g_{Fe})$)	Selectivity (%C)				O/P ratio of C_2–C_4	CO_2 (% of CO converted)	α
			CH_4	C_2–C_4 olefins	C_2–C_4 paraffins	C_{5+}			
0	49.3	7.0	18.0	28.9	16.3	36.7	1.8	39.2	0.58
0.001	39.6	5.6	18.9	26.5	20.7	33.9	1.3	32.2	0.57
0.002	26.4	3.7	20.6	26.4	23.0	29.7	1.1	24.6	0.54
0.005	17.4	2.5	17.8	26.7	24.1	31.5	1.1	17.4	0.52
0.01	5.6	0.8	17.2	25.8	29.5	27.6	0.9	13.0	0.48
0.02	–	–	–	–	–	–	–	–	–

Note: Reaction conditions: H_2/CO = 1, 2.0 MPa, 24000 mL/(h g_{cat}), 573 K, TOS = 7 h.
'–' indicates almost no activity.

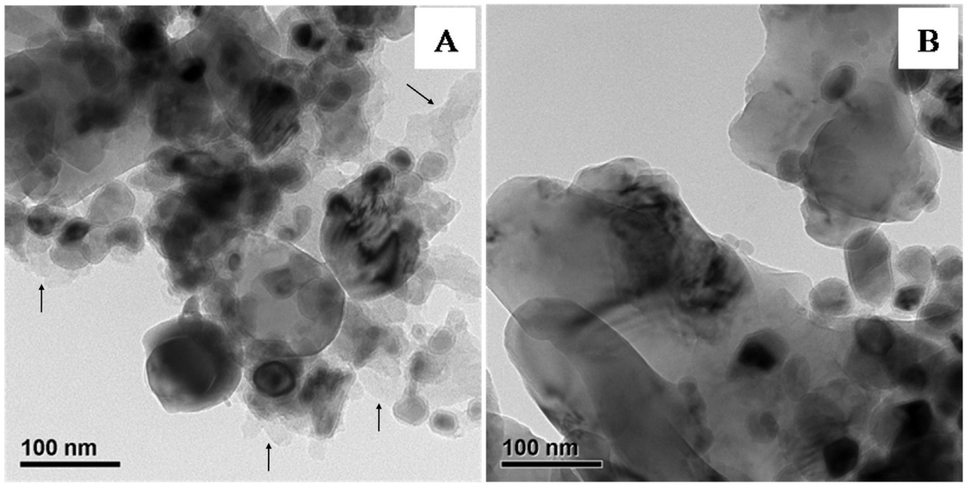

Fig. 10. TEM images of the (A) Fe/α-Al_2O_3 and (B) $S_{0.01}$Fe/α-Al_2O_3 catalysts after FTS reaction. (2.0 MPa, H_2/CO = 1, 573 K).

0.001 and 0.01 decreases the CO conversion. In the case of $S_{0.02}$Fe/α-Al_2O_3 catalyst, the FTS activity almost diminished to zero. This result indicates that, for the FTS reaction at 573 K, a small amount of sulfur (S/Fe = 0.02) was sufficient to completely deactivate the Fe/α-Al_2O_3 catalyst. As indicated in the XPS spectra of the catalysts after the FTS reaction (Fig. 8), both sulfate and sulfur species were detected on the $S_{0.05}$Fe/α-Al_2O_3. It has been shown that the nature of sulfur species (S^{2-}, $SO_4{}^{2-}$) would affect catalyst behavior for the FTS reaction [13,23]. A low-level sulfide iron catalyst with sulfate species on the surface was associated with improving the FTS activity, whereas the presence of S^{2-} ions at the surface showed effects of poisoning [13]. It was also found that, at low sulfur coverage, each sulfide atom was able to poison 8–10 atoms of iron [62]. Based on these results, it can be concluded that the sulfide species is responsible for the lower FTS activity of the sulfur modified Fe/α-Al_2O_3 catalysts as compared with the sulfur free one. The poisoning effect of sulfur on the Fe/α-Al_2O_3 catalyst may have resulted from the blockage of catalytic metal sites that prevent the adsorption and activation of reactant molecules. As shown in Table 2, the presence of sulfur also had a profound impact on the product distribution of the FTS reaction. With increasing the S/Fe molar ratios in the catalysts from 0 to 0.01, the selectivity toward C_{5+} hydrocarbons and the chain growth probabilities (α) of the catalysts continuously decreased from 36.7 to 27.6 % and 0.58 to 0.48, respectively, whereas the total selectivity to C_2–C_4 hydrocarbons increased gradually from 45.2 to 55.3 %. Comparatively, CH_4 selectivity of the catalysts was not strongly affected by sulfur. As shown in Fig. S4, product distribution over Fe/α-Al_2O_3 catalyst follows the ASF model, whereas CH_4 selectivities over $S_{0.005}$Fe/α-Al_2O_3 and $S_{0.01}$Fe/α-Al_2O_3 catalysts are below the ASF prediction, indicating that addition of sulfur resulted in a deviation of CH_4 selectivity from the ASF model. It is also worth to note that sulfur-modification decreases the olefin-to-paraffin (O/P) ratio of C_2–C_4 hydrocarbons in the reaction products. With increasing of the sulfur content (S/Fe molar ratio) in the catalysts from 0.001 to 0.01, the selectivity of C_2–C_4 olefins remained almost unchanged, whereas the selective of C_2–C_4 paraffins increased from 20.7 to 29.5%. This phenomenon can be well explained by that sulfur modification increasing the H/C ratio on the catalyst surface. As indicated in the H_2-TPD (Fig. 6) and CO-TPD (Fig. 7) profiles of the catalysts, the intensity of the CO desorption peak with T_m > 600 K corresponding to the dissociative adsorption of CO decreased with increasing of sulfur content, whereas the intensity of the H_2 desorption peak at temperature of about 500 K increased slightly with increasing of sulfur content. This will lead to an enhanced hydrogenation ability of the carbon species on the catalyst surface. As a result, the chain growth termination by hydrogen and the olefin hydrogenation are promoted. The hydrocarbon products shift to short-chain molecules containing more paraffins. Higher H/C ratio leads to lower C_{5+} selectivity and O/P ratio are well documented in the literature [63].

Fig. 10 shows the TEM image of the Fe/α-Al_2O_3 and $S_{0.01}$Fe/α-Al_2O_3 catalysts after the FTS reaction at 573 K and 2.0 MPa. Comparing with the particle size of H_2-reduced samples (Fig. 2), a significant increase in the average size of iron-containing particles was observed on the spent Fe/α-Al_2O_3 (25 ± 7 nm) and $S_{0.01}$Fe/α-Al_2O_3 (30 ± 10 nm) catalysts. It can also be seen that the surface of spent Fe/α-Al_2O_3 catalyst (Fig. 10A) is covered with large amounts of amorphous carbon (pointed by arrows). Comparatively, carbon deposition on the $S_{0.01}$Fe/α-Al_2O_3 catalyst (Fig. 10B) is negligible. This indicates that the addition of sulfur decreased the formation of amorphous carbon. The study of Torres Galvis et al. also indicated

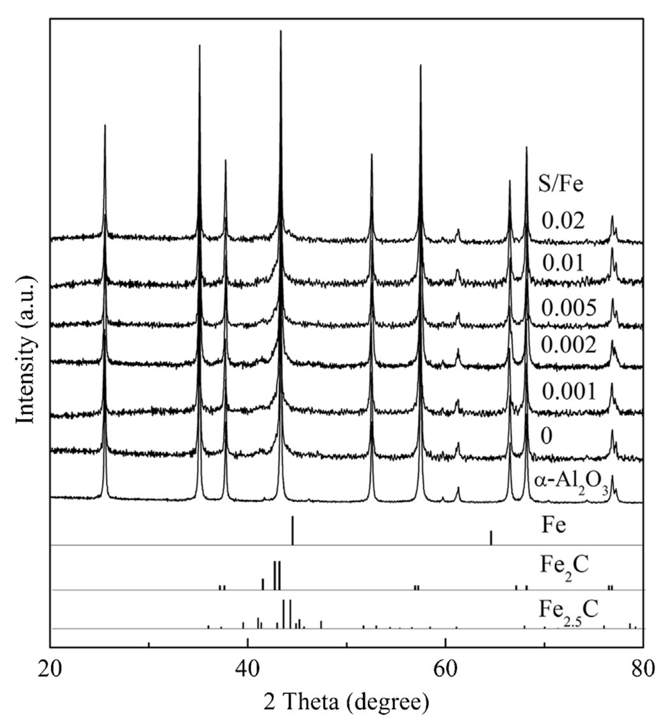

Fig. 11. XRD patterns of the S_xFe/α-Al_2O_3 catalysts after the FTS reaction. (H_2/CO = 1, 2.0 MPa, 573 K).

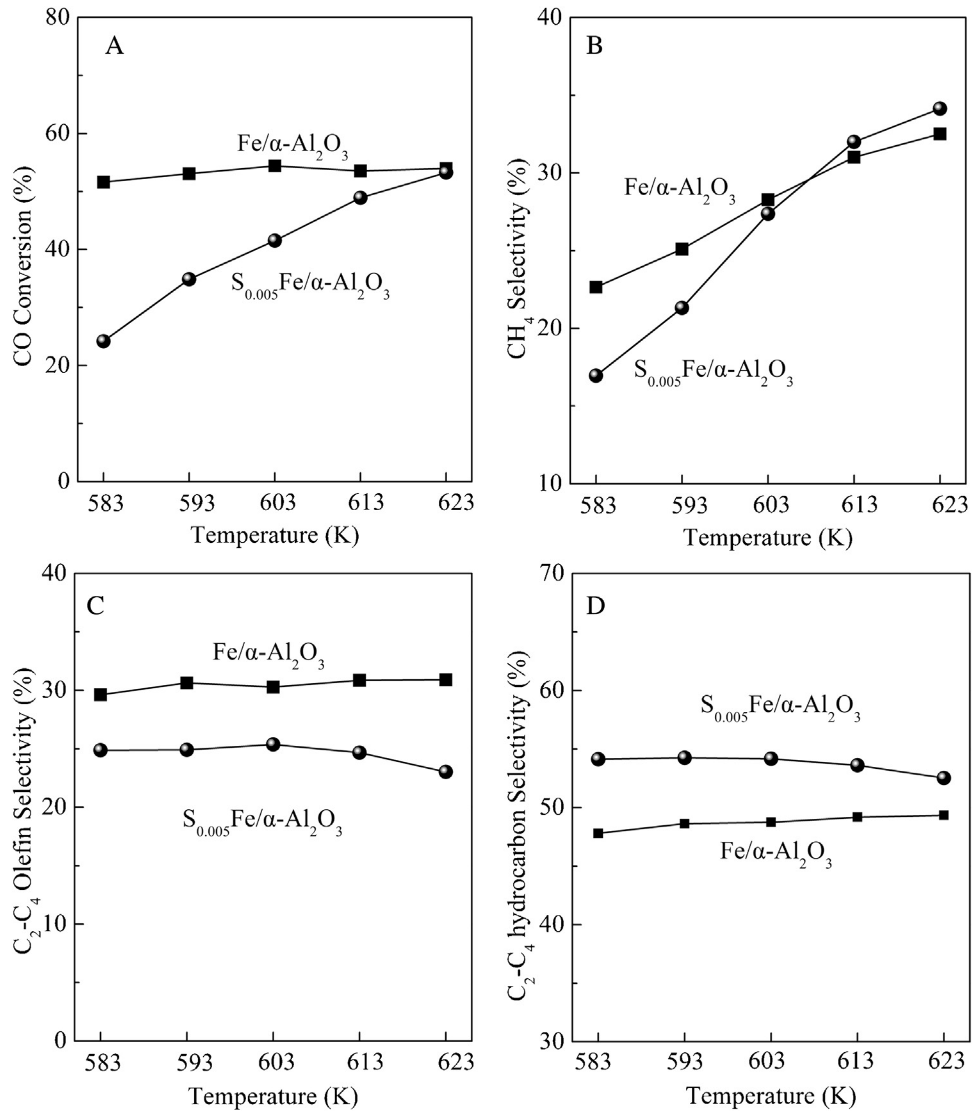

Fig. 12. Effect of reaction temperature on the (A) CO conversion, (B) CH_4 selectivity, (C) C_2–C_4 olefin selectivity and (D) C_2–C_4 hydrocarbon selectivity of the Fe/α-Al_2O_3 and $S_{0.005}$Fe/α-Al_2O_3 catalysts. (Reaction conditions: H_2/CO = 1, 2.0 MPa, 24000 mL/(h g_{cat}), TOS = 3 h.)

that the catalysts modified with sulfur show a low amount of carbon deposits [16]. This observation can also be related to suppression of dissociative activation of CO species by sulfur. XRD patterns of the Fe/α-Al_2O_3 and S_xFe/α-Al_2O_3 (x = 0.001–0.02) catalysts after FTS reaction are displayed in Fig. 11. It can be seen that XRD peaks attributed to iron carbides (Fe_xC_y) were observed on all catalysts, and no diffraction peak referring to metallic iron or iron oxides was detected. This observation is different from the results of XPS analysis of the CO/H_2 pretreated $S_{0.05}$Fe/α-Al_2O_3 sample (Fig. S3). The underlying reason for the difference may be that the sample for XRD analysis was treated under high pressure condition which speeded up the carbonization of metallic iron species in the catalyst. Since most intense peak of iron carbides partly overlaps with the peak of α-Al_2O_3 at about 43.5°, the type of iron carbides species formed on the catalysts can not be clearly identified by XRD. Nevertheless, the XRD patterns clearly indicate that the intensities of the diffraction peak of iron carbides (especially the peaks for Fe_2C) on the used catalysts gradually decrease with the increasing of S/Fe molar ratios. Bromfield et al. reported a similar observation for a precipitated iron catalyst modified by Na_2S [13]. This phenomenon can be understood in terms of sulfur suppressing the dissociative adsorption of CO, and thereby inhibiting the carburization of iron species. Under FTS reaction conditions, the iron-based catalyst is known to be composed of many phases, including α-Fe, Fe_3O_4 and iron carbides [64]. Although there is considerable controversy concerning of the catalytically active iron phase as well as the role of each iron carbide species in the FTS reaction, the catalysts with more iron carbide species were often found to exhibit high activity for the FTS reaction [65–68]. This suggests that lower CO conversions observed on the S_xFe/α-Al_2O_3 as compared with that on Fe/α-Al_2O_3 may have resulted from the suppression of carburization of iron species by sulfur under FTS reaction conditions. As indicated in the CO-TPD profiles (Fig. 7) of the S_xFe/α-Al_2O_3 catalysts, addition of sulfur significantly decreases the amount of dissociative adsorbed CO on the Fe/α-Al_2O_3 catalyst, especially on the samples with S/Fe molar ratio above 0.01. This will result in a decrease in the concentration of active carbon species (C_{ad}) available for the formation of iron carbides and hydrocarbons, and eventually lead to a decline in FTS activity of the catalysts.

Fig. 12 shows the effect of reaction temperature on FTS performance of the Fe/α-Al_2O_3 and $S_{0.005}$Fe/α-Al_2O_3 catalysts. It can be seen that the addition of sulfur decreases the CO conversion of Fe/α-Al_2O_3 catalyst at low reaction temperature. As the reaction temperature was increased from 583 to 623 K, CO conversion of

$Fe/\alpha-Al_2O_3$ catalyst remains almost unchanged, while CO conversion of $S_{0.005}Fe/\alpha-Al_2O_3$ catalyst increases significantly. At 623 K, CO conversion of $S_{0.005}Fe/\alpha-Al_2O_3$ catalyst almost equals to that of $Fe/\alpha-Al_2O_3$. This observation indicates that the poisoning effect of sulfur on $Fe/\alpha-Al_2O_3$ catalyst is more significant at low reaction temperature. The significant difference in the dependence of activity (CO conversion) of $Fe/\alpha-Al_2O_3$ and $S_{0.005}Fe/\alpha-Al_2O_3$ catalysts on the reaction temperature could be related to the difference in the extent of carbon formation over the catalysts under FTS reaction conditions. As indicated in the TEM image of the spent $Fe/\alpha-Al_2O_3$ catalyst shown in Fig. 10, severe carbon deposition occurred on the $Fe/\alpha-Al_2O_3$ catalyst after the FTS reaction. It is well known that at high temperature, carbon formation by the Boudouard reaction will be more facile [69]. If the carbon species are not removed by hydrogenation to form hydrocarbons, they can block the active catalytic sites and eventually cause breakup of catalyst particles when they form carbon filaments [70–73]. Comparatively, the extent of carbon deposition on the sulfur modified $Fe/\alpha-Al_2O_3$ under FTS reaction conditions is much less significant (9 10). It could be the reason why the activity (CO conversion) of sulfur free catalyst is almost not affected by the variation of temperature, whereas the activity of sulfur modified catalysts increased significantly with increasing reaction temperature. This result indicates that the activity of iron catalyst poisoned by sulfur depends on temperature and thus the increasing of reaction temperature can improve the $Fe/\alpha-Al_2O_3$ catalyst's resistance to sulfur poisoning.

As also indicated in Fig. 12, the CH_4 selectivity of $Fe/\alpha-Al_2O_3$ and $S_{0.005}Fe/\alpha-Al_2O_3$ catalysts increased greatly with the increasing of reaction temperature. $S_{0.005}Fe/\alpha-Al_2O_3$ catalyst exhibited similar or slightly higher CH_4 selectivity in comparison with $Fe/\alpha-Al_2O_3$ catalyst at similar CO conversion level at 623 K. It indicates that, under the reaction conditions of this study, addition of sulfur does not reduce CH_4 selectivity of the catalysts. Within a temperature range between 583 and 623 K, the $S_{0.005}Fe/\alpha-Al_2O_3$ catalyst exhibited lower $C_2–C_4$ olefin selectivity and higher $C_2–C_4$ hydrocarbon selectivity as compared with $Fe/\alpha-Al_2O_3$ catalyst. These results indicate that both high reaction temperature and sulfur modification are in favor of the formation of lighter hydrocarbons. However, the olefin to paraffin (O/P) ratio of the $C_2–C_4$ fraction of $S_{0.005}Fe/\alpha-Al_2O_3$ is lower than that of $Fe/\alpha-Al_2O_3$, leading to a decrease in olefin selectivity over the sulfur modified catalyst.

4. Conclusions

The effects of sulfur (S/Fe = 0.001 ∼ 0.05, molar ratio) on $Fe/\alpha-Al_2O_3$ catalysts for FTS were systemically studied. The catalysts were characterized by TEM, (*in situ*) XRD, XPS, and TPR/TPD. The results show that incorporation of sulfur into a $Fe/\alpha-Al_2O_3$ catalyst promoted the reduction of FeO to metallic Fe during the H_2 reduction process. With increasing the sulfur content, the rates of reduction, carburization and carbon deposition of the catalysts under CO atmosphere decreased. Sulfide (S^{2-}) and sulfate species(SO_4^{2-})are the main sulfur components in the $S_xFe/\alpha-Al_2O_3$ catalysts after H_2 reduction and FTS reaction. The sulfide species on the H_2-reduced $S_xFe/\alpha-Al_2O_3$ catalysts are located on the surface of metallic Fe particles. A small amount of sulfur could enhance the amount of H_2 desorption from the catalyst surface at a temperature of about 500 K. However, an excess of sulfur on the catalyst suppressed the H_2 adsorption. The dissociative adsorption of CO on the H_2-reduced $Fe/\alpha-Al_2O_3$ catalysts was retarded by the addition of sulfur. The presence of sulfur inhibited the formation of iron carbides and amorphous carbon during FTS reaction. FTS performance of $S_xFe/\alpha-Al_2O_3$ depended on the content of sulfur as well as on the reaction temperature. The negative effect of sulfur on the FTS activity of $Fe/\alpha-Al_2O_3$ catalyst can be related to the inhibition of CO dissociation and iron carbide formation over the catalyst under the FTS conditions (2.0 MPa, H_2/CO = 1, 573 K). Resistance to sulfur poisoning of the $Fe/\alpha-Al_2O_3$ catalyst was found to improve with increasing reaction temperature. The presence of sulfur in the catalyst also resulted in a decrease in selectivity of C_{5+} hydrocarbons as well as olefin to paraffin (O/P) ratio of $C_2–C_4$ products and promoted the formation of $C_2–C_4$ hydrocarbons. The higher surface H/C ratio over the sulfur modified catalyst under FTS conditions was proposed to be the main factor responsible for the changes in the product distribution.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.apcata.2015.12.023.

References

[1] Q.H. Zhang, J.C. Kang, Y. Wang, ChemCatChem 2 (2010) 1030–1058.
[2] H.M. Torres Galvis, K.P. de Jong, ACS Catal. 3 (2013) 2130–2149.
[3] M. Janardanarao, Ind. Eng. Chem. Res. 29 (1990) 1735–1753.
[4] R.J. Madon, H. Shaw, Catal. Rev. Sci. Eng. 15 (1977) 69–106.
[5] J.F. Shultz, L.J.E. Hofer, F.S. Karn, R.B. Anderson, J. Phys. Chem. 66 (1962) 501–506.
[6] J. Bayer, K.C. Stein, L.J.E. Hofer, R.B. Anderson, J. Catal. 3 (1964) 145–155.
[7] F.S. Karn, J.F. Shultz, R.E. Kelly, R.B. Anderson, I&EC Prod. Res. Dev. 3 (1964) 33–38.
[8] R.B. Anderson, F.S. Karn, J.F. Shultz, J. Catal. 4 (1965) 56–63.
[9] P.K. Agrawal, W.D. Fitzharris, J.R. Katzer, Stud. Surf. Sci. Catal. 6 (1980) 179–200.
[10] J. Barrault, C. Forquy, V. Perrichon, Appl. Catal. 5 (1983) 119–125.
[11] C.H. Bartholomew, R.M. Bowman, Appl. Catal. 15 (1985) 59–67.
[12] Z.T. Liu, J.L. Zhou, B.J. Zhang, J. Mol. Catal. 94 (1994) 255–261.
[13] T.C. Bromfield, N.J. Coville, Appl. Catal. A: Gen. 186 (1999) 297–307.
[14] J.A. Kritzinger, Catal. Today 71 (2002) 307–318.
[15] T.Z. Li, Y. Yang, Z.C. Tao, H.J. Wan, X. An, C.H. Zhang, H.W. Xiang, Y.W. Li, J. Nat. Gas Chem. 16 (2007) 354–362.
[16] H.M. Torres Galvis, A.C.J. Koeken, J.H. Bitter, T. Davidian, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, J. Catal. 303 (2013) 22–30.
[17] A.J. McCue, J.A. Anderson, Catal. Sci. Technol. 4 (2014) 272–294.
[18] H.G. Stenger, C.N. Satterfield, I&EC Process Des. Dev. 24 (1985) 415–420.
[19] Z.T. Liu, Y.W. Li, J.L. Zhou, Z.X. Zhang, B.J. Zhang, Appl. Catal. A: Gen. 161 (1997) 137–151.
[20] B. Wu, L. Tian, L. Bai, Z. Zhang, H. Xiang, Y.W. Li, Catal. Commun. 5 (2004) 253–257.
[21] B.S. Wu, L. Bai, H.W. Xiang, Y.W. Li, Z.X. Zhang, B. Zhong, Fuel 83 (2004) 205–212.
[22] W.L. van Dijk, J.W. Niemantsverdriet, A.M. van dar Kraan, H.S. van der Baan, Appl. Catal. 2 (1982) 273–288.
[23] T.C. Bromfield, N.J. Coville, Appl. Surf. Sci. 119 (1997) 19–24.
[24] H.M. Torres Galvis, A.C.J. Koekena, J.H. Bitter, T. Davidian, M. Ruitenbeekb, A.I. Dugulanc, K.P. de Jong, Catal. Today 215 (2013) 95–102.
[25] H.M. Torres Galvis, J.H. Bitter, C.B. Khare, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, Science 335 (2012) 835–838.
[26] C.D. Wagner, L.E. Davis, M.V. Zeller, J.A. Taylor, R.H. Raymond, L.H. Gale, Surf. Interface Anal. 3 (1981) 211–225.
[27] T.R. Motjope, H.T. Dlamini, G.R. Hearne, N.J. Coville, Catal. Today 71 (2002) 335–341.
[28] J. Xu, C.H. Bartholomew, J. Phys. Chem. B 109 (2005) 2392–2403.
[29] Z.H. Chonco, A. Ferreira, L. Lodya, M. Claeys, E. van Steen, J. Catal. 307 (2013) 283–294.
[30] T.R. Motjope, T.H. Dlamini, AIP Conf. Proc. 765 (2005) 43–48.
[31] H. Wang, Y. Yang, B.S. Wu, J. Xu, M.Y. Ding, H.L. Wang, W.H. Fan, H.W. Xiang, Y.W. Li, J. Mol. Catal. A: Chem. 308 (2009) 96–107.
[32] S. Li, A. Li, S. Krishnamoorthy, E. Iglesia, Catal. Lett. 77 (2001) 197–205.
[33] X. Cui, J. Xu, C. Zhang, Y. Yang, P. Gao, B. Wu, Y. Li, J. Catal. 282 (2011) 35–46.
[34] M.C. Ribeiro, G. Jacobs, B.H. Davis, D.C. Cronauer, A.J. Kropf, C.L. Marshall, J. Phys. Chem. C 114 (2010) 7895–7903.

[35] J. Benziger, R.J. Madix, Surf. Sci. 94 (1980) 119–153.
[36] D.W. Moon, D.J. Dwyer, S.L. Bernasek, Surf. Sci. 163 (1985) 215–229.
[37] D.W. Moon, S.L. Bernasek, J.P. Lu, J.L. Gland, D.J. Dwyer, Surf. Sci. 184 (1987) 90–108.
[38] F. Bozso, G. Ertl, M. Grunze, M. Weiss, Appl. Surf. Sci. 1 (1977) 103–119.
[39] P.B. Merrill, R.J. Madix, Surf. Sci. 347 (1996) 249–264.
[40] H. Ahlafi, D. Bianchi, C.O. Bennett, Appl. Catal. 66 (1990) 99–110.
[41] J. Li, C. Zhang, X. Cheng, M. Qing, J. Xu, B. Wu, Y. Yang, Y. Li, Appl. Catal. A: Gen. 464–465 (2013) 10–19.
[42] G.D. Weatherbee, J.L. Rankin, C.H. Bartholomew, Appl. Catal. 11 (1984) 73–84.
[43] Y. Amenomiya, G. Pleizier, J. Catal. 28 (1973) 442–454.
[44] H.Z. Liu, C.B. Liu, X.N. Li, Y.Q. Cen, Ind. Eng. Chem. Res. 42 (2003) 1347–1349.
[45] W. Arabczyk, I. Jasińska, R. Pelka, Catal. Today 169 (2011) 97–101.
[46] G. Brodén, G. Gafner, H.P. Bonzel, Surf. Sci. 84 (1979) 295–314.
[47] U. Seip, I.C. Bassignana, J. Küppers, G. Ertl, Surf. Sci. 160 (1985) 400–418.
[48] U. Seip, M.C. Tsai, K. Christmann, J. Küppers, G. Ertl, Surf. Sci. 139 (1984) 29–42.
[49] J.L. Rankin, C.H. Bartholomew, J. Catal. 100 (1986) 533–540.
[50] J. Oh, T. Kondo, D. Hatake, K. Arakawa, T. Suzuki, D. Sekiba, J. Nakamura, J. Phys. Chem. C 116 (2012) 7741–7747.
[51] G. Brodén, G. Gafner, H.P. Bonzel, Appl. Phys. 13 (1977) 333–342.
[52] T.N. Rhodin, C.F. Brucker, Solid State Commun. 23 (1977) 275–279.
[53] F. Jona, K.O. Legg, H.D. Shih, D.W. Jepsen, P.M. Marcus, Phys. Rev. Lett. 40 (1978) 1466–1469.
[54] K.O. Legg, F. Jona, D.W. Jepsen, P.M. Marcus, Surf. Sci. 66 (1977) 25–37.
[55] T. Jin, M. Machida, T. Yamaguchi, K. Tanabe, Inorg. Chem. 23 (1984) 4396–4398.
[56] E. Paparazzo, J. Electron. Spectrosc. Relat. Phenom. 43 (1987) 97–112.
[57] C.S. Kuivila, J.B. Butt, P.C. Stair, Appl. Surf. Sci. 32 (1988) 99–121.
[58] D. Brion, Appl. Surf. Sci. 5 (1980) 133–152.
[59] E. de Smit, M.M. van Schooneveld, F. Cinquini, H. Bluhm, P. Sautet, F.M.F. de Groot, B.M. Weckhuysen, Angew. Chem. Int. Ed. 50 (2011) 1584–1588.
[60] H.P. Bonzel, G. Brodén, H.J. Krebs, Appl. Surf. Sci. 16 (1983) 373–394.
[61] T. Grzybek, H. Papp, M. Baerns, Appl. Catal. 29 (1987) 351–359.
[62] C.H. Bartholomew, P.K. Agrawal, J.R. Katzer, Adv. Catal. 31 (1982) 135–242.
[63] G.P. Van der Laan, A. Beenackers, Catal. Rev. Sci. Eng. 41 (1999) 255–318.
[64] E. de Smit, B.M. Weckhuysen, Chem. Soc. Rev. 37 (2008) 2758–2781.
[65] W. Chen, Z.L. Fan, X.L. Pan, X.H. Bao, J. Am. Chem. Soc. 130 (2008) 9414–9419.
[66] M.D. Shroff, D.S. Kalakkad, K.E. Coulter, S.D. Kohler, M.S. Harrington, N.B. Jackson, A.G. Sault, A.K. Datye, J. Catal. 156 (1995) 185–207.
[67] J.D. Xu, K.T. Zhu, X.F. Weng, W.Z. Weng, C.J. Huang, H.L. Wan, Catal. Today 215 (2013) 86–94.
[68] J.A. Amelse, J.B. Butt, L.H. Schwartz, J. Phys. Chem. 82 (1978) 558–563.
[69] A.C.J. Koeken, H.M. Torres Galvis, T. Davidian, M. Ruitenbeek, K.P. de Jong, Angew. Chem. Int. Ed. 51 (2012) 7190–7193.
[70] M.E. Dry, A.P. Steynberg, Stud. Surf. Sci. Catal. 152 (2004) 406–481.
[71] C.H. Bartholomew, Appl. Catal. A: Gen. 212 (2001) 17–60.
[72] K.P. De Jong, J.W. Geus, Catal. Rev. 42 (2000) 481–510.
[73] J.R. Rostrup-Nielsen, J. Sehested, Adv. Catal. 47 (2002) 65–139.